

gemstone-rs Explorer Guide

Local HTTP explorer endpoints and safety defaults



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Explorer Guide

`gemstone-rs-explorer` is a local-only HTTP explorer built on the Rust API. It is useful for proving the browser, inspect, eval, and codegen surfaces before embedding richer tooling in VS Code or another frontend.

Install and Run

```
cargo install gemstone-rs-explorer
gemstone-rs-explorer --port 8787
```

From a checkout:

```
cargo run -p gemstone-rs-explorer -- --port 8787
```

Open:

```
http://127.0.0.1:8787/
```

Screenshot:

gemstone-rs Explorer

Local-only GemStone/S explorer.

- [/api/config](#)
- [/api/status](#)
- [/api/browse/dictionaries](#)
- [/api/browse/classes?dictionary=UserGlobals](#)
- [/api/browse/protocols?class=Object](#)
- [/api/browse/methods?class=Object&protocol=- all -](#)
- [/api/browse/source?class=Object](#)
- [/api/codegen/sample](#)
- [/api/codegen/preview?config=examples/codegen/gemstone-rs.codegen](#)
- [/api/codegen/diff?config=examples/codegen/gemstone-rs.codegen](#)
- [/api/codegen/check?config=examples/codegen/gemstone-rs.codegen](#)
- [/api/inspect?oop=20](#)

read_only=true allow_eval=false

Safety Defaults

The explorer:

- binds to `127.0.0.1` by default
- rejects non-loopback hosts
- starts read-only
- requires `--allow-eval` before workspace evaluation
- requires `--allow-write` before codegen writes
- reads GemStone credentials from the same `GS_*` environment as the CLI

Do not expose it publicly without adding authentication and transport security.

Browse Endpoints

Run these from a second terminal:

```
curl -s http://127.0.0.1:8787/health
curl -s http://127.0.0.1:8787/api/config
curl -s http://127.0.0.1:8787/api/doctor
curl -s 'http://127.0.0.1:8787/api/doctor?live=1'
```

```
curl -s http://127.0.0.1:8787/api/status
curl -s http://127.0.0.1:8787/api/browse/dictionaries
curl -s 'http://127.0.0.1:8787/api/browse/classes?dictionary=UserGlobals'
curl -s 'http://127.0.0.1:8787/api/browse/protocols?class=Object&meta=0'
curl -s 'http://127.0.0.1:8787/api/browse/methods?class=Object&protocol=--%20all%20--&meta=0'
curl -s 'http://127.0.0.1:8787/api/browse/source?class=Object&selector=printString&meta=0'
curl -s 'http://127.0.0.1:8787/api/inspect?oop=20'
```

Expected shapes:

```
{"status":"ok"}
{"host":"127.0.0.1","port":8787,"readOnly":true,"allowEval":false}
{"success":true,"environment":{"GS_PASSWORD":{"status":"set","masked":true}}}
{"connected":true,"sessionId":12345,"needsCommit":false,"inTransaction":false}
{"success":true,"dictionaries":["UserGlobals"]}
```

Eval

Eval is disabled by default:

```
gemstone-rs-explorer --allow-eval
```

Then:

```
curl -s 'http://127.0.0.1:8787/api/eval?source=3%20%2B%204'
```

Expected:

```
{"type":"smallInt","value":7}
```

Codegen Endpoints

Read-only endpoints:

```
curl -s http://127.0.0.1:8787/api/codegen/sample
curl -s 'http://127.0.0.1:8787/api/codegen/preview?config=examples/codegen/gemstone-rs.codegen'
curl -s 'http://127.0.0.1:8787/api/codegen/diff?config=examples/codegen/gemstone-rs.codegen'
curl -s 'http://127.0.0.1:8787/api/codegen/check?config=examples/codegen/gemstone-rs.codegen'
```

```
curl -s 'http://127.0.0.1:8787/api/codegen/discover-mapping?
mapped=BookingDraft&class=Object'
```

Expected check result when generated output is current:

```
{"success":true,"exists":true,"upToDate":true}
```

Write-gated endpoint:

```
gemstone-rs-explorer --allow-write
```

```
curl -s 'http://127.0.0.1:8787/api/codegen/generate?config=examples/codegen/gemstone-
rs.codegen'
```

Expected:

```
{"success":true,"output":"examples/codegen/generated/
gemstone_wrappers.rs","bytes":1234}
```

BridgeRoot Endpoints

BridgeRoot inspection and mapping-config preview use the same live session configuration as the browser endpoints:

```
curl -s http://127.0.0.1:8787/api/bridge/root
curl -s http://127.0.0.1:8787/api/bridge/keys
curl -s 'http://127.0.0.1:8787/api/bridge/get?key=BookingDraft'
curl -s 'http://127.0.0.1:8787/api/bridge/get?key=BookingDraft&key_type=Symbol'
curl -s 'http://127.0.0.1:8787/api/bridge/mapping-config?mapped=BookingDraft'
```

Writes are disabled unless the explorer is started with `--allow-write`:

```
gemstone-rs-explorer --allow-write
curl -s 'http://127.0.0.1:8787/api/bridge/put?key=WorkbenchDraft&value=hello'
curl -s 'http://127.0.0.1:8787/api/bridge/put?
key=WorkbenchCount&value=7&value_type=SmallInt'
curl -s 'http://127.0.0.1:8787/api/bridge/put?
key=WorkbenchApproved&value=true&value_type=Bool'
curl -s 'http://127.0.0.1:8787/api/bridge/remove?key=WorkbenchDraft'
```

Expected shapes:

```
{"success":true,"name":"GemStoneRsBridgeRoot","oop":1234,"identityId":1}
{"success":true,"root":"GemStoneRsBridgeRoot","keys":
[{"printString":"BookingDraft"]]}
{"oop":5678,"classOop":9012,"printString":"aDictionary(...)" }
{"success":true,"config":"mapped = BookingDraft ..."}
{"success":true,"root":"GemStoneRsBridgeRoot","key":"WorkbenchDraft","keyType":"String",
"valueType":"String","oop":3456}
```

Product Direction

The explorer should remain a separate binary crate. The core `gemstone-rs` library stays focused on safe GemStone API access, while the explorer proves higher-level browsing and codegen workflows.

Good next steps:

- richer frontend over the current HTTP endpoints
- generated wrapper preview and file diff UI
- explicit codegen selection config
- optional VS Code webview embedding

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